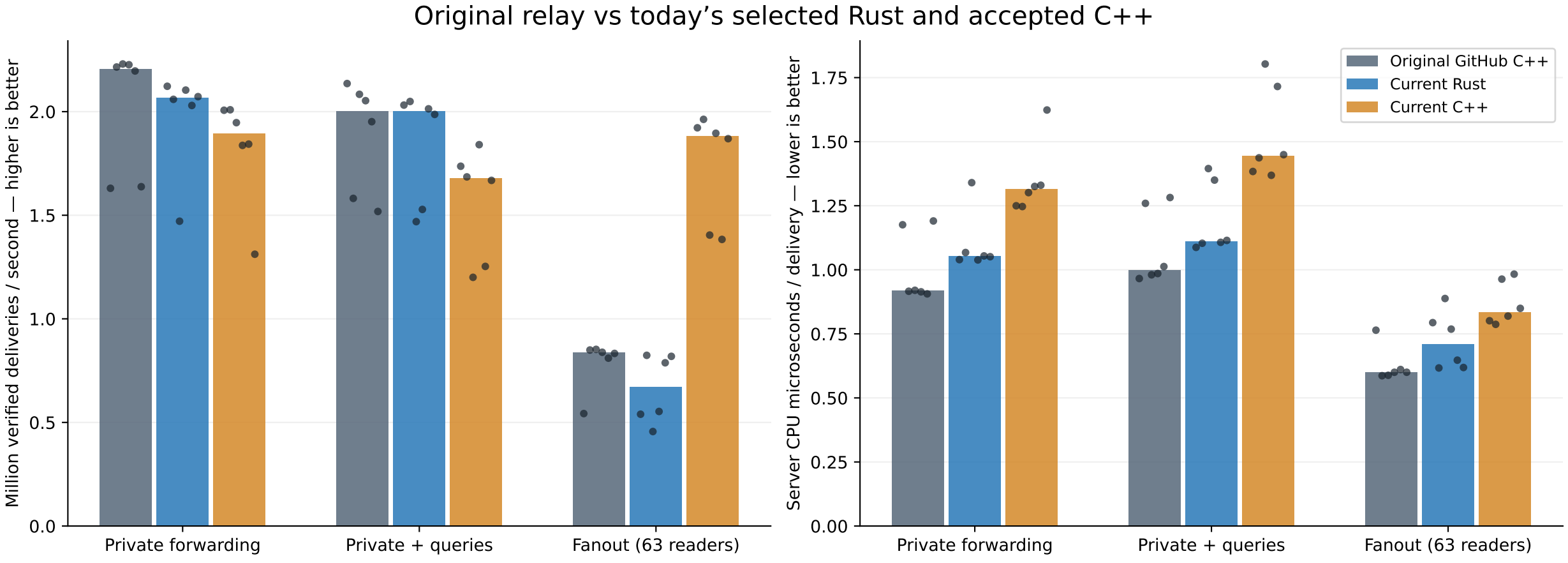
Original relay vs today’s selected Rust and accepted C++
Million verified deliveries / second — higher is better
0.0
0.5
1.0
1.5
2.0
Private forwarding
Private + queries
Fanout (63 readers)
Server CPU microseconds / delivery — lower is better
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Private forwarding
Private + queries
Fanout (63 readers)
Original GitHub C++
Current Rust
Current C++
Bars: medians of 6 fresh trials. Dots: all repeats. 4 server cores · 1 KiB values · 8,192 idle rooms
Timed workload only; exact payload/order checked. Native protocols; no bridge. Loopback, 6 seconds/trial.
Query: one Rust repeat missed 1 of 1,200 scheduled queries; its timed data is retained.